Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

20.1 Money Supply

Meaning of money supply

- Money supply refers to the amount of money stock of an economy.

Money supply = Cash in public circulation + Bank deposits

Definitions of money supply in Hong Kong

- M1 includes only those things generally accepted as a medium of exchange in society.

M1 = Legal tender notes and coins held by the public + Customers' demand deposits with licensed banks

M1 only includes the legal tender notes and coins held by the public, those held by authorised institutions are not included.

- M2 and M3 emphasise the function of money as a store of value.

M2 = M1 + Customers' savings and time deposits with licensed banks + Negotiable certificates of deposit (NCDs) issued by licensed banks held outside the banking sector

Remittance abroad is money no longer circulated in Hong Kong and is thus not included in any of the definitions of Hong Kong money supply.

M3 = M2 + Customers' deposits with restricted licence banks and deposit-taking companies + NCDs issued by restricted licence banks and deposit-taking companies held outside the banking sector

20.2 Deposit Creation and Deposit Contraction

Deposit creation of commercial banks

- Fractional reserve system
  - Under the fractional reserve system, deposit-taking institutions are subject to reserve requirements on customers' deposits.

$$\text{Required reserve ratio} = \frac{\text{Required reserves}}{\text{Total deposits}} \times 100\%$$

Required reserves = Total deposits × Required reserve ratio

Excess reserves = Actual reserves – Required reserves

When banks do not hold excess reserves, actual reserves will be equal to required reserves.

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- Assumptions for calculating the maximum change in deposits:
  - **The banking system holds no excess reserves.**
  - **There is no cash leakage in the banking system.**
- Process of **deposit creation**

  Suppose someone deposits an amount of cash in the banking system.

  → The **reserves increase by the amount deposited**, but the required reserves increase by only a fraction of the amount deposited. There will be **excess reserves in the banking system**.

  → The banks will **lend out the excess reserves**.

  → The **bank loans will be re-deposited into the banking system**.

  The process will go on and on until the actual reserves are equal to the required reserves.
- After the process of deposit creation is completed:

$$\text{Maximum change in deposits} = \text{Initial change in deposits} \times \frac{1}{\text{Required reserve ratio}}$$

$$\text{Maximum change in loans} = \text{Initial change in loans} \times \frac{1}{\text{Required reserve ratio}}$$

$$\text{Change in reserves} = \text{Change in deposits} - \text{Change in loans}$$

$$\text{Maximum change in money supply} = \text{Change in cash in public circulation} + \text{Maximum change in bank deposits}$$

Changes in deposits, loans, reserves and money supply after the process of deposit contraction is completed can also be calculated using these formulae.

Deposit contraction of commercial banks

- Assumptions for calculating the maximum change in deposits:
  - The banking system does not have excess reserves.
  - After the initial withdrawal, the amount of cash the public hold does not change further.

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- Process of **deposit contraction**

  Suppose someone withdraws an amount of cash from the banking system.

  → **Actual reserves** held by the banking system **are not sufficient to fulfil the minimum reserve requirement**.

  → Banks have to **call back some of the loans** from debtors or sell some of their assets to the public in order to fulfil the reserve requirement.

  → The public have to **withdraw deposits from their accounts** to pay back the loans or purchase the assets of banks.

  The process of deposit contraction will go on and on until the actual reserves are equal to the required reserves.

When there is a change in deposits, the resulted change in money supply is not necessarily equal to the change in deposits.

Maximum banking multiplier and actual banking multiplier

- **Maximum banking multiplier**

$$\text{Maximum banking multiplier} = \frac{1}{\text{Required reserve ratio}}$$

$$\text{Maximum change in deposits} = \text{Initial change in deposits} \times \text{Maximum banking multiplier}$$

- **Actual banking multiplier**

$$\text{Actual reserve ratio} = \frac{\text{Actual reserves}}{\text{Total deposits}} \times 100\%$$

$$= \frac{\text{Required reserves} + \text{Excess reserves}}{\text{Total deposits}} \times 100\%$$

$$\text{Actual banking multiplier} = \frac{1}{\text{Actual reserve ratio}}$$

- When banks hold excess reserves,
  - the actual reserve ratio is higher than the required reserve ratio.
  - the actual banking multiplier is smaller than the maximum banking multiplier.
- Limitations of deposit creation
  - cash leakage
  - excess reserves

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Money supply and monetary base

When the required reserve ratio decreases, the monetary base will decrease.

$$\text{Monetary base} = \text{Cash in public circulation} + \text{Bank reserves}$$

$$\text{Money multiplier} = \frac{\text{Money supply}}{\text{Monetary base}}$$

- Even though the government can precisely control the size of the monetary base, it cannot fully control the amount of money supply, which depends on:
  - the currency deposit ratio; and
  - the reserve deposit ratio/the actual reserve ratio.

20.3 Money Demand

Purposes of holding money

- **Money demand** is the total amount of money which the public desire to hold.
- **Transaction demand for money**
  - It refers to the amount of money which the public desire to hold for daily transaction of goods and services.
  - It is positively related to nominal income.
- Asset **demand for money**
  - It refers to the amount of money the public desire to hold as an asset for the purpose of storing wealth.
  - The nominal interest rate is the opportunity cost of holding money. **When people choose to hold money, they forgo the return on holding other interest-bearing assets.**
  - The quantity of money held as an asset is negatively related to the nominal interest rate.
- **Total demand for money**
  - The total demand for money is the sum of the transaction demand for money and the asset demand for money.
- Change in total quantity demanded of money versus change in demand for money
  - Change in total quantity demanded of money is shown by a movement along the same money demand curve.
  - Change in demand for money is shown by a shift of the money demand curve.

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Factors affecting demand for money

- Transaction demand for money ↑ (↓) when:
  - nominal income, which is affected by total amount of output and general price level, increases (decreases).
  - the use of advanced payment methods becomes less (more) popular.
  - the time interval between income payments increases (decreases).
  - people's desire to consume increases (decreases).
- Asset demand for money ↑ (↓) when:
  - people's wealth increases (decreases).
  - the risks of holding other assets increases (decreases).
  - the liquidity of other interest-bearing assets decreases (increases).



A change in inflationary expectation will lead to a change in the nominal interest rate. The quantity demanded of money instead of the demand for money will change.

20.4 Determination of the Equilibrium Interest Rate in the Money Market

The importance of interest rate in an economy

- An increase (a decrease) in the interest rate will lower (raise) the levels of planned investment and planned consumption, which will eventually lower (raise) the levels of national income and employment in an economy.

Determination of the equilibrium interest rate

- The **money supply curve** is vertical.
- The **money demand curve** is downward sloping.
- The intersection of the money demand curve (M_d) and the money supply curve (M_S) determines the **equilibrium interest rate** (r_e).

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Changes in the equilibrium interest rate

- Changes in money supply
  - When money supply increases (decreases), the equilibrium interest rate will decrease (increase).
- Changes in money demand
  - When money demand increases (decreases), the equilibrium interest rate will increase (decrease).
- Changes in both money supply and money demand

Money demand	Extent	Money supply	Nominal interest rate
↑	>	↑	↑
	<		↓
	=		remains unchanged
↓	>	↓	↓
	<		↑
	=		remains unchanged
↑	>	↓	↑
	<		
	=		
↓	>	↑	↓
	<		
	=		

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